Intro to Randomized Algorithms

Introduction to Randomized Algorithms

1. A Quick Review of Probability

2. The Hiring Problem

3. Generating a Random Permutation

4. Various Other Items
   - Shuffling Cards
   - The Birthday Paradox
   - Coupon Collectors
   - Generating Random Numbers

A quick review of probability theory

A discrete random variable takes on a countable number of distinct values (usually counts).

Expectation. The *expectation*, $E[X]$, of a discrete random variable X, is defined as:

$$E[X] = \sum i \cdot \Pr[X = i]$$

Q: Roll a 6-sided dice. What is the expected value?

A: $E[X] = \sum_{i=1}^{6} i \cdot \frac{1}{6} = 3.5$

Q: Roll two dice. What is the expected TOTAL value?

A: $E[X] = 2 \cdot \frac{1}{36} + 3 \cdot \frac{2}{36} + 4 \cdot \frac{3}{36} + 5 \cdot \frac{4}{36} + 6 \cdot \frac{5}{36} + 7 \cdot \frac{6}{36} + 8 \cdot \frac{5}{36} + 9 \cdot \frac{4}{36} + 10 \cdot \frac{3}{36} + 11 \cdot \frac{2}{36} + 12 \cdot \frac{1}{36} = 7$

Q: Roll two dice. What is the expected MAXIMUM value seen on a die?

A: $E[X] = 1 \cdot \frac{1}{36} + 2 \cdot \frac{3}{36} + 3 \cdot \frac{5}{36} + 4 \cdot \frac{7}{36} + 5 \cdot \frac{9}{36} + 6 \cdot \frac{11}{36} = 4.47$

A quick review of probability theory (cont)

Expectation. The *expectation*, $E[X]$, of a discrete random variable X, is defined as:

$$E[X] = \sum i \cdot \Pr[X = i]$$



Q (waiting time for the first success): Coin comes up heads with probability p and tails with probability $1 - p$.
How many flips X until first head is seen (lets call it the waiting time, for later use).

A:

$$E[X] = \sum_{j=1}^{\infty} j \cdot Pr[X = j] = \sum_{j=1}^{\infty} j \cdot (1-p)^{j-1} p = p \sum_{j=1}^{\infty} j \cdot (1-p)^{j-1} = p \cdot \frac{1}{p^2} = \frac{1}{p}$$

$j - 1$ tails 1 head

waiting time=$1/p$

Exercise on a new series

Show that for $0 < p < 1$: $\sum_{j=1}^{\infty} j(1-p)^{j-1} = \frac{1}{p^2}$

Hint: use the infinite geometric series, for $0 < x < 1$: $\sum_{k=0}^{\infty} x^k = 1 + x + x^2 + \ \ldots = \frac{1}{1-x}$

The solution is similar to the solution for the geometric series.

Set $x = 1 - p$ $(x < 1)$. Then, $S = \sum_{j=1}^{\infty} j(1-p)^{j-1} = \sum_{j=1}^{\infty} jx^{j-1} = \sum_{k=0}^{\infty} (k+1)x^k = 1 + 2x + 3x^2 + 4x^3 + \cdots$

Multiply with x: $xS = \sum_{j=1}^{\infty} jx^j = \sum_{k=0}^{\infty} (k+1)x^{k+1} = x + 2x^2 + 3x^3 + 4x^4 + \cdots$

Compute $S - xS$: $S - xS = (1-x)S = 1 + x + x^2 + x^3 + \cdots = \frac{1}{1-x}$

Thus: $S = \frac{1}{(1-x)^2} = \frac{1}{p^2}$

Linearity of expectation

(1). Given two random variables X and Y (not necessarily independent),

$$E[X+Y] = E[X] + E[Y].$$

Remark: $E[XY] = E[X]E[Y]$ *only when* X *and* Y *are independent.*

Example: Roll two dice. What is the expected TOTAL value X?

We saw that we can calculate

$$E[X] = 2\cdot\frac{1}{36} + 3\cdot\frac{2}{36} + 4\cdot\frac{3}{36} + 5\cdot\frac{4}{36} + 6\cdot\frac{5}{36} + 7\cdot\frac{6}{36} + 8\cdot\frac{5}{36} + 9\cdot\frac{4}{36} + 10\cdot\frac{3}{36} + 11\cdot\frac{2}{36} + 12\cdot\frac{1}{36} = 7$$

Easier way is to let X_1, X_2 be the random variables that are values of first and second die. Then $E[X_1] = 3.5, E[X_2] = 3.5$ and by *Linearity of Expectation*

$$E[X_1+X_2] = E[X_1] + E[X_2] = 3.5 + 3.5 = 7.$$

Exercise Coupon Collector

Coupon collector. Each box of cereal contains a coupon. There are n different types of coupons. Assuming a box contains each type of coupon equally likely, how many boxes do you need to open to have at least one coupon of each type?

Solution.

- Stage i = number of boxes to open between having i and $i+1$ distinct coupons.
- What is the probability p of success at stage 0?
- 1
- What is the probability p of success at stage i?
- $p = \frac{n-i}{n}$
- Let X_i = number of boxes to open at stage i (waiting time at stage i)
- What is the expected value of X_i ?
- $\frac{1}{p} = \frac{n}{n-i}$
- What is the expected number of boxes you are expected to open in total?

$$E[X] = \sum_{i=0}^{n-1} E[X_i] = \sum_{i=0}^{n-1} \frac{n}{n-i} = n \sum_{i=1}^{n} \frac{1}{i} = \Theta(n \log n) \qquad (\because \sum_{i=1}^{n} 1/i = \Theta(\log n))$$

Indicator random variables

An indicator random variable X only takes values 0 or 1: $E[X] = \Pr[X = 1]$.

Example. Shuffle a deck of n cards; turn them over one at a time; try to guess each card (n guesses in total). Assume you can't remember what's already been turned over and just guess a card from full deck uniformly at random.

Q. What is the expected number of **correct** guesses?

A. (surprisingly effortless using linearity of expectation)

- Let $X_i = 1$ if i^{th} guess is correct and 0 otherwise.
- Let X = number of correct guesses $= X_1 + \cdots + X_n$.

- $E[X_i] = \Pr[X_i = 1] = 1/n$.
- $E[X] = E[X_1] + \cdots + E[X_n] = 1/n + \cdots + 1/n = 1$.

Exercise on Guessing Cards *with Memory*

Guessing with memory. Shuffle a deck of n cards; turn them over one at a time; try to guess each card. You remember all the cards that have been turned over.

Q. What's the expected number of correct guesses?

A.

- Let $X_i = 1$ if i^{th} guess is correct and 0 otherwise.
- Let X = number of correct guesses $= X_1 + \cdots + X_n$.

- $E[X_i] = \Pr[X_i = 1] = 1/(n - i + 1)$.

- $E[X] = E[X_1] + \cdots + E[X_n] = \frac{1}{n} + \cdots + \frac{1}{2} + \frac{1}{1} = \Theta(\log n)$.

Will now use this to analyze a simple RANDOMIZED algorithm

The Hiring Problem

```
Hire-Assistant(n):
best ← 0
for i ← 1 to n
    interview candidate i
    if candidate i is better than best then
        hire candidate i
        best ← i
```

Q: How many people are hired in the worst case and when;

Q: How to avoid the randomness of the input

A: Insert randomness in the algorithm itself

The Hiring Problem: Randomized Algorithm

```
Hire-Assistant(n):
randomly permute all n candidates
best ← 0
for i ← 1 to n
    interview candidate i
    if candidate i is better than best then
        hire candidate i
        best ← i
```

Q: What is the expected number of hires?

A: Similar to *Guessing Cards with Memory*

- Let $X_i = 1$ if you hire candidate i and 0 otherwise.
- Set $X =$ number of hires $= X_1 + \cdots + X_n$.

- $E[X_i] = \Pr[X_i = 1] = 1/i$. (Among the first i candidates, the best has probability $1/i$ to be placed at the last position.)

- $E[X] = E[X_1] + \cdots + E[X_n] = 1 + \frac{1}{2} + \cdots + \frac{1}{n-1} + \frac{1}{n} = \Theta(\log n)$.